UNIT 4 – SCIENCE & NATURE

WARM-UP: SCIENCE AROUND US!

TASK 1: EVERYDAY SCIENCE & NATURE'S INVENTIONS

Part A – Nature did it first!

Many modern inventions were actually copied from nature. Work with a partner and match each invention (1–5) with its natural inspiration (A–E).

Modern Invention	Match	Inspiration from Nature
1. Airplanes		A. Bat's sound signals
2. Velcro		B. Bird's wings
3. Solar panels		C. Lotus leaves
4. Waterproof clothes		D. Plant leaves
5. Radar & Parking sensors		E. Sticky plant seeds

Part B – Fact or Fiction? (True or False)

*We hear a lot of "science tips" on TikTok or from our grandparents. Guess if these statements are **True (T)** or **False (F)**.*

Statement	True (T)	False (F)
1. Putting a wet phone in a bowl of uncooked rice helps fix it.		
2. Chameleons change color mainly to express their mood, not just to hide.		
3. Eating carrots helps you see clearly in total darkness.		
4. Lightning is actually hotter than the surface of the Sun.		
5. Listening to Mozart while sleeping makes you wake up smarter.		
6. Bananas are slightly radioactive but completely safe to eat.		

Part C – Quick Chat (2 minutes)

1. Which answer in Part A or Part B surprised you the most?
2. Have you ever believed a "fake science tip" on social media? What was it?

TASK 2: HUMAN VS. NATURE - FIND THE BALANCE

Part A – Positive or Negative Impact?

*Look at the human activities below. Put a **(+)** if it helps nature, and a **(-)** if it harms nature. Can you add one more idea to each side?*

- [ ] **1.** Using AI cameras to track and protect endangered tigers.
- [ ] **2.** Cutting down forests to build resorts and highways.
- [ ] **3.** Testing cosmetics and shampoos on lab animals.
- [ ] **4.** Growing meat in a laboratory so we don't have to kill cows.
- [ ] **5.** Releasing plastic waste and chemicals into the ocean.
- [ ] **6.** Creating biodegradable bags from cassava and corn.

Part B – "Would You Rather?"

Ask your partner to choose ONE option in each pair and explain why in 1–2 sentences.

1. **Option 1:** Would you rather **travel to Mars** on a spaceship OR **explore the deepest ocean** on Earth?
2. **Option 2:** If you were a billionaire, would you spend money to **bring dinosaurs back to life** OR **clean up all plastic in the ocean**?
3. **Option 3:** Would you rather live in a **100% smart high-tech city** (with no trees) OR a **green forest village** (with weak Wi-Fi)?

TASK 3: SPEAKING TOOLBOX – SHARE YOUR VIEW!

Pick ONE question from Activity 2, interview your partner, and share their answer with the class using the useful phrases below.

Useful Vocabulary:

- **learn from nature** *(học hỏi từ thiên nhiên)*
- **protect endangered species** *(bảo vệ các loài có nguy cơ tuyệt chủng)*
- **eco-friendly inventions** *(các phát minh thân thiện với môi trường)*
- **do more harm than good** *(lợi bất cập hại / hại nhiều hơn lợi)*
- **improve our quality of life** *(nâng cao chất lượng cuộc sống)*
- **keep the natural balance** *(giữ cân bằng tự nhiên)*

Sentence Starters:

- *"In Activity 1, my partner and I were really surprised to learn that..."*
- *"When asked whether to choose [A] or [B], my partner chose... because..."*
- *"In my opinion, science is great, ___ut we must be careful not to..."*

LISTENING

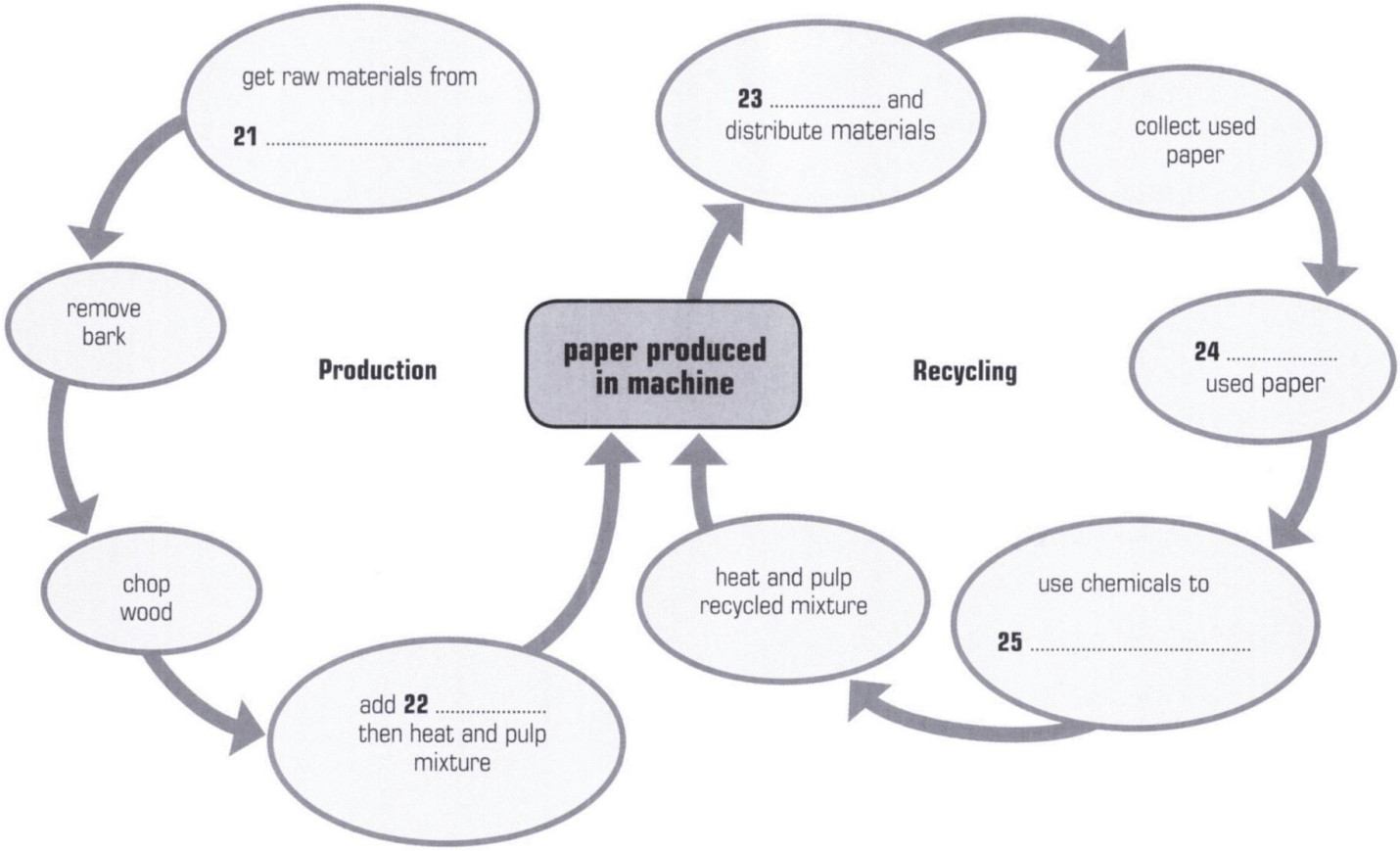
FLOW-CHART COMPLETION

PRACTICE 1 (Track 4.1)

Questions 21 - 25

Complete the flow-chart below.

Write **NO MORE THAN TWO WORDS** for each answer.



Questions 26 - 30

Answer the questions below.

Write NO MORE THAN TWO WORDS for each answer.

26. What part of the assignment is Alan going to start working on? _____

27. Where will Melanie get more information on used paper collection? _____

28. What will they add to the assignment to make it more interesting? _____

29. What do they agree to complete by the end of the month? _____

30. Who will they ask to review their work? _____

PRACTICE 2 (Track 4.2)

Questions 21–25

Choose the correct letter**, A, B, or C.**

Science experiment for Year 12 students

1. How does Clare feel about the students in her Year 12 science class?

A. Worried that they are not making progress
B. Challenged by their poor behaviour in class
C. Frustrated at their lack of interest in the subject

2. How does Jake react to Clare's suggestion about an experiment based on children's diet?

A. He is concerned that the results might not be meaningful.
B. He feels some of the data might be difficult to obtain.
C. He suspects that the conclusions might be upsetting.

3. What problem do they agree may be involved in an experiment involving animals?

A. Any results may not apply to humans.
B. It may be complicated to get permission.
C. Students may not be happy about animal experiments.

4. What question do they decide the experiment should address?

A. Are mice capable of controlling their food intake?
B. Does an increase in sugar lead to health problems?
C. How much do supplements of different kinds affect health?

5. Clare might also consider doing another experiment involving:

A. Other types of food supplement.
B. Different genetic strains of mice.
C. Varying amounts of exercise.

Questions 26–30

Complete the flowchart below.

Choose **FIVE** answers from the box and write the correct letter, **A–H,** next to Questions **26–30.**

A Size
B Escape
C Age
D Water
E Cereal
F Calculations
G Changes
H Colour

Choose mice which are all the same **26** _____

↓

Divide the mice into two groups, each with a different **27** _____

↓

Put each group in a separate cage.
Feed group A commercial mouse food.
Feed group B the same, but also sugar contained in **28** _____

↓

Take measurements using an electronic scale.
Place them in a weighing chamber to prevent **29** _____

↓

Do all necessary **30** _____

READING

SUMMARY / FLOW-CHART COMPLETION

PRACTICE 1

The problems and benefits created by the spread of the water hyacinth in Kenya

Water hyacinth (Eichhornia crassipes), an aquatic plant native to South America, first appeared in countries in Africa in the early 1900s. Scientists there called it the 'world's worst aquatic weed', after it spread from the southernmost tip of Africa in the early 1900s and started obstructing major dams and rivers.

In east Africa the plant arrived with Belgian colonists in Rwanda, who liked the look of its glossy leaves and delicate purple flowers floating in their ponds. But by the 1980s, it had 'escaped' out of the country via the Kagera river and made its way downstream to Lake Victoria. There, with no natural predators and perfect temperature conditions, the plant began spreading in the open water, blocking fishing routes and providing a new habitat for disease-carrying mosquitoes.

For the women who smoke fish from the lake to sell it has meant declining income, as the boats that once brought the fish to shore by the hundreds struggle to navigate through the mass of plants. But water hyacinth isn't their only headache. In order to smoke the fish that they buy, they must gather huge quantities of firewood, sometimes walking as far as 10km each way to collect enough to complete their work. And each day as they cook, they breathe in the thick, grey smoke. About three out of four families in Kenya depend on wood or charcoal to cook their daily meals, and the rate is even higher in rural areas, Kenya's latest demographic and health survey shows. Using solid fuels like these for cooking increases indoor pollution. The World Health Organization estimates that about 14,300 Kenyans die annually as a result of indoor air pollution - most of which is caused by cooking and heating sources.

Some years ago, on the shores of Lake Victoria, huge piles of water hyacinth that villagers had taken out of the water in an attempt to clear it were a common sight. But buried in those decaying waxy leaves was a renewable energy gold mine. It turns out the floating plant isn't just good at spreading - its foliage also contains a high ratio of carbon to nitrogen. It's a magic combination that has captivated researchers' imaginations since as early as the 1980s when, across the world, they began to explore its potential as a biofuel. Just about 4kg of the dried plant would be enough to cater for a large family's daily energy needs, early research predicted.

In 2014, Nigerian academics announced they had got better yields of biofuel gas when they mixed the plant with chicken manure. A few years later, Kenyan scientists confirmed what their Nigerian peers and others had already found: manure worked to improve the process of converting the weed into gas.

In 2018, the technology came to a village on the shore of Lake Victoria, called Dunga. The project promised a two-for-one solution to the dual menaces of the water hyacinth and dependence on firewood. The community received a pair of donated biogas digesters - machines that would transform a mix of water hyacinth and cow dung into biogas for cooking.

The digesters work a bit like a stomach. The mixture goes in one end - think of it as a mouth - and over the next 20 to 30 days, it goes through a fermentation process and breaks down, giving off gas that comes out the other end. From there, the clean burning gas is passed through pipes to the point of use, just like traditional domestic gas. In Dunga, the machines produce enough gas to serve about 60% of the village's population. It is used in domestic stoves and for other household tasks such as purifying water and incubating chicks.

The project is testing whether biogas can provide an effective alternative to firewood and charcoal in rural Kenyan communities. Results indicate that the programme seems to be working. The women who smoke the lake fish are already getting sick less often. Besides, they don't have to devote a lot of time every day to gathering firewood, which is a great relief. As a result, they're able to make more money for their families from other enterprises.

Kanyiva Muindi is an epidemiologist and air pollution research fellow at the African Population and Health Research Centre in Nairobi. She says families who switch to the smokeless cooking method could expect fewer respiratory diseases. Women, young girls and children are particularly vulnerable because they are the ones who cook in the kitchen or outside over fires.

How much better the biogas stoves will be for the community's health still needs more research, says Dominic Kahumbu Wanjihia, Biogas International's chief executive. But unless the price of the machines drops, it's pretty clear that most communities will never be able to afford any, since they sell for about $750.

Kanyiva says affordability is a challenge worth addressing, given the huge health and environmental dangers posed by 'dirty' fuels such as wood, charcoal and kerosene. If biogas could become affordable on a large scale, she says it 'would be life-changing for millions on the African continent and beyond'.

Questions 1-7

Do the following statements agree with the information given in the Reading Passage?

Write **TRUE if the statement agrees with the information**
FALSE if the statement contradicts the information
NOT GIVEN if there is no information on this

1 Water hyacinth was introduced as a decorative plant in east Africa.
2 Fishermen took some water hyacinth plants to Lake Victoria.
3 It is now difficult to force boats through the thick water hyacinth on Lake Victoria.
4 Chemicals produced by the water hyacinth plants are affecting the numbers of fish in Lake Victoria.
5 Cooking with charcoal has been proved to be even worse for people's health than cooking with wood.
6 People found it impossible to remove much water hyacinth from Lake Victoria.
7 Scientists started investigating the possibility of using water hyacinth to generate biogas in the last century.

Questions 8-10

Complete the flow-chart below.

Choose NO MORE THAN TWO WORDS from the passage for each answer.

Generating biogas for domestic use in Dunga

First, place water hyacinth together with some **8** ..... into a digester.

↓

Leave the mixture until the **9** ..... is completed

↓

Capture the gas emitted by the digester and use **10** ..... to transport it to individual homes

↓

Then use the gas for cooking as well as making water fit for human consumption

Questions 11-13

Complete the notes below.

Choose ONE WORD ONLY from the passage for each answer.

Cooking with biogas in Dunga

Benefits for the women in the village of cooking with biogas

- no need for them to spend so much **11** ..... collecting fuel
- they can focus on different tasks that bring in **12** .....
- they are less likely to experience certain diseases connected to burning wood

Drawbacks of changing to biogas

- the **13** ..... of the digesters is beyond the reach of most villages

PRACTICE 2

What is exploration?

We are all explores. Our desire to discover, and then share that new-found knowledge, is part of what makes us human – indeed, this has played an important part in our success as a species. Long before the first caveman slumped down beside the fire and grunted news that there were plenty of wildebeest over yonder, our ancestors had learnt the value of sending out scouts to investigate the unknown. This questing nature of ours undoubtedly helped our species spread around the globe, just as it nowadays no doubt helps the last nomadic Penan maintain their existence in the depleted forests of Borneo, and a visitor negotiate the subways of New York.

Over the years, we've come to think of explorers as a peculiar breed – different from the rest of us, different from those of us who are merely 'well travelled', even; and perhaps there is a type of person more suited to seeking out the new, a type of caveman more inclined to risk venturing out. That, however, doesn't take away from the fact that we all have this enquiring instinct, even today; and that in all sorts of professions – whether artist, marine biologist or astronomer – borders of the unknown are being tested each day.

Thomas Hardy set some of his novels in Egdon Heath, a fictional area of uncultivated land, and used the landscape to suggest the desires and fears of his characters. He is delving into matters we all recognise because they are common to humanity. This is surely an act of exploration, and into a world as remote as the author chooses. Explorer and travel writer Peter Fleming talks of the moment when the explorer returns to the existence he has left behind with his loved ones. The traveller 'who has for weeks or months seen himself only as a puny and irrelevant alien crawling laboriously over a country in which he has no roots and no background, suddenly encounters his other self, a relatively solid figure, with a place in the minds of certain people'.

In this book about the exploration of the earth's surface, I have confined myself to those whose travels were real and who also aimed at more than personal discovery. But that still left me with another problem: the word 'explorer' has become associated with a past era. We think back to a golden age, as if exploration peaked somehow in the 19th century – as if the process of discovery is now on the decline, though the truth is that we have named only one and a half million of this planet's species, and there may be more than 10 million – and that's not including bacteria. We have studied only 5 per cent of the species we know. We have scarcely mapped the ocean floors, and know even less about ourselves; we fully understand the workings of only 10 per cent of our brains.

Here is how some of today's 'explorers' define the word. Ran Fiennes, dubbed the 'greatest living explorer', said, 'An explorer is someone who has done something that no human has done before – and also done something scientifically useful. ' Chris Bonington, a leading mountaineer, felt exploration was to be found in the act of physically touching the unknown: 'You have to have gone somewhere new. ' Then Robin Hanbury-Tenison, a campaigner on behalf of remote so-called 'tribal' peoples, said, 'A traveller simply records information about some far-off world, and reports back; but an explorer changes the world. ' Wilfred Thesiger, who crossed Arabia's Empty Quarter in 1946, and belongs to an era of unmechanised travel now lost to the rest of us, told me, 'If I'd gone across by camel when I could have gone by car, it would have been a stunt. ' To him, exploration meant bringing back information from a remote place regardless of any great self-discovery.

Each definition is slightly different – and tends to reflect the field of endeavour of each pioneer. It was the same whoever I asked: the prominent historian would say exploration was a thing of the past, the cutting-edge scientist would say it was of the present. And so on. They each set their own particular criteria; the common factor in their approach being that they all had, unlike many of us who simply enjoy travel or discovering new things, both a very definite objective from the outset and also a desire to record their findings.

I'd best declare my own bias. As a writer, I'm interested in the exploration of ideas. I've done a great many expeditions and each one was unique. I've lived for months alone with isolated groups of people all around the

world, even two 'uncontacted tribes'. But of these things is of the slightest interest to anyone unless, through my books, I've found a new slant, explored a new idea. Why? Because the world has moved on. The time has long passed for the great continental voyages – another walk to the poles, another crossing of the Empty Quarter. We know how the land surface of our planet lies; exploration of it is now down to the details – the habits of microbes, say, or the grazing behaviour of buffalo. Aside from the deep sea and deep underground, it's the era of specialists. However, this is to disregard the role the human mind has in conveying remote places; and this is what interests me: how a fresh interpretation, even of a well-travelled route, can give its readers new insights.

Questions 27-32

Choose the correct letter, A, B, C or D.

27. The writer refers to visitors to New York to illustrate the point that

A exploration is an intrinsic element of being human.
B most people are enthusiastic about exploring.
C exploration can lead to surprising results.
D most people find exploration daunting.

28. According to the second paragraph, what is the writer's view of explorers?

A Their discoveries have brought both benefits and disadvantages.
B Their main value is in teaching others.
C They act on an urge that is common to everyone.
D They tend to be more attracted to certain professions than to others.

29. The writer refers to a description of Egdon Heath to suggest that

A Hardy was writing about his own experience of exploration.
B Hardy was mistaken about the nature of exploration.
C Hardy's aim was to investigate people's emotional states.
D Hardy's aim was to show the attraction of isolation.

30. In the fourth paragraph, the writer refers to 'a golden age' to suggest that

A the amount of useful information produced by exploration has decreased.
B fewer people are interested in exploring than in the 19th century.
C recent developments have made exploration less exciting.
D we are wrong to think that exploration is no longer necessary.

31. In the sixth paragraph, when discussing the definition of exploration, the writer argues that

A people tend to relate exploration to their own professional interests.
B certain people are likely to misunderstand the nature of exploration.
C the generally accepted definition has changed over time.
D historians and scientists have more valid definitions than the general public.

32. In the last paragraph, the writer explains that he is interested in

A how someone's personality is reflected in their choice of places to visit.
B the human ability to cast new light on places that may be familiar.
C how travel writing has evolved to meet changing demands.
D the feelings that writers develop about the places that they explore.

Questions 33 - 37

Look at the following statements (Questions 33 - 37) and the list of explorers below. Match each statement with the correct explorer, A-E. You may use any letter more than once.

33 He referred to the relevance of the form of transport used.

34 He described feelings on coming back home after a long journey.
35 He worked for the benefit of specific groups of people.
36 He did not consider learning about oneself an essential part of exploration.
37 He defined exploration as being both unique and of value to others.

Lists of Explorers		
A Peter Fleming	**C** Chris Bonington	**E** Wilfred Thesiger
B Ran Fiennes	**D** Robin Hanbury-Tenison	

Questions 38 - 40
Complete the summary below.
Choose NO MORE THAN TWO WORDS from the passage for each answer.

The writer's own bias

The writer has experience of a large number of **38** _____ , and was the first stranger that certain previously **39** _____ people had encountered. He believes there is no need for further exploration of Earth's **40** _____, except to answer specific questions such as how buffalo eat.

SPEAKING

PRACTICE 1

Part 1:

1. Did you enjoy studying science subjects when you were in secondary school? Why (not)?
2. Do you think you use scientific knowledge in your daily life? In what ways?
3. What is a scientific discovery or invention that you find particularly fascinating?
4. How do you keep up with news about scientific breakthroughs or technological trends?
5. Do you prefer watching documentaries about the universe or about biology on Earth?

Part 2:

Describe an interesting science experiment or demonstration you saw or took part in at school.
You should say

- What the experiment was about
- When and where you saw or did it
- What equipment or materials were used
- And explain why this experiment left a strong impression on you.

Part 3:

1. Why do many high school students find science subjects like Physics or Chemistry theoretical and hard to understand?
2. How are AI tools and smart apps changing the way teenagers do their science homework today?
3. Do you think schools should encourage more girls to pursue STEM careers in the future?

PRACTICE 2

Part 1:

1. How often do you spend time in natural outdoor environments like parks or botanical gardens?
2. What kind of natural scenery do you find most relaxing when you feel stressed?
3. Have you ever planted a tree or grown vegetables at home? How was the experience?
4. Is there any wild animal native to your country that you are particularly fond of?
5. Do you think big cities nowadays provide enough green spaces for young residents?

Part 2:

Describe a natural place you visited that had a lot of plastic waste or environmental pollution.
You should say:

- Where this place was and when you went there
- Why you decided to visit this place
- What kind of pollution or damage you noticed
- And explain how you felt about what you saw.

Part 3

1. Why do young people continue to use single-use plastics like milk tea cups and straws despite knowing they harm nature?
2. Do you think teenagers in crowded cities like Hanoi or Ho Chi Minh City have enough access to green parks and nature?
3. Should high schools organize mandatory outdoor clean-up trips or tree-planting activities for students?

WRITING

TASK 1 – PROCESS

I. Writing a report for task 1: PROCESSES

1. Procedure for Processes:

- Identify how many steps / stages there are in total.
- Look at the ingredients / materials / tools needed in each step.

2. Sentence structures:

- Paraphrase sentence: The diagram explains how …
- Overview sentence: Overall, a total of … stages are involved in the production, commencing with ... and ending with .... The entire process also + [Extra information about the WHOLE process: time, use of tools, one single location]

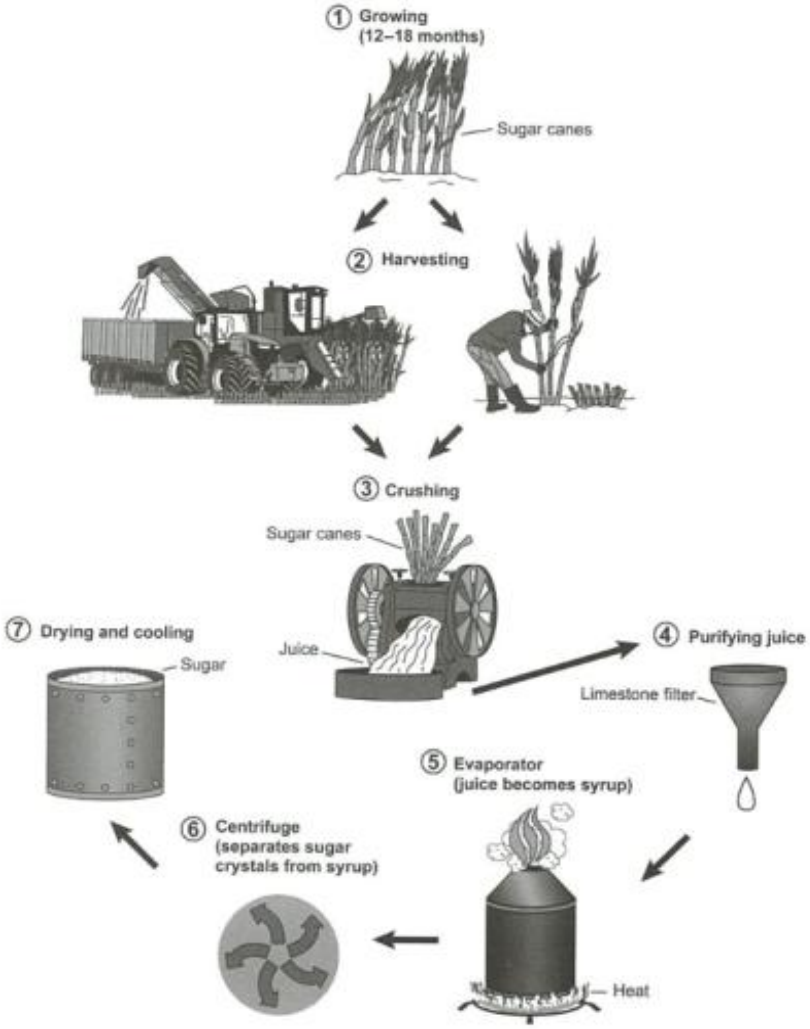
II. Practice:

PRACTICE 1

You should spend about 20 minutes on this task.

The diagram below shows the manufacturing process for making sugar from sugar cane. Summarize the information by selecting and reporting the main features, and make comparisons where relevant.

Write at least 150 words.



1. Look at the Diagram. Answer the questions below.

a. What process does the map describe? _____

b. Is it natural or man-made? _____

c. Which voice should be used? active / passive

2. Complete the Introduction sentence with the answers above.

The diagram illustrates the _____ process of _____
The diagram illustrates how _____ **(is produced / produces) (artificially / naturally)** from sugar canes.

3. Choose the words in brackets that best describe the main features of the diagram.

Overall, a total of _____ steps are involved in the **(linear / cyclical)** process, beginning with **(harvesting mature sugar canes / sugar canes reaching maturity / fully developed sugar canes)** and ending with **(a container full of sugar / sugar being dried and cooled / the processed sugar)**. It is also quite technically **(sophisticated / simple / undemanding)** despite requiring only one ingredient.

4. Look at the outline below. Fill the blanks with words or phrases in the box.

fed	**industrial reaper**	**impurities**	**manually**
sugar canes	**metal canister**	**rotates**	**heated**
juice	**sugar**	**left**	**purified**

When?	**What?**	**What is done?**	**Where / When / How / Why / Into what?**
The process starts with **(production / cultivation / germination)**	_____ _____	planted	on farms for 12 to 18 months
		harvested	either by an _____ _____or _____
During the **(processing / evaporation / purification)** stage		_____	into a machine which crushes them into _____
	The juice	passes	through a filter made of limestone where its _____ are removed
	The _____ juice	_____	inside an evaporator until it becomes a thick syrup
	The syrup	tranferred	into a centrifuge which _____ at a very high speed so that sugar crystals are separated from the liquid
	The _____	_____	inside a _____ _____ to dry and cool
	This completes the production **(sequence / loop)**		

5. The "When" column tells us why these are the main stages (groups of steps) of the production diagrams. Choose the correct words in brackets to complete the outline.

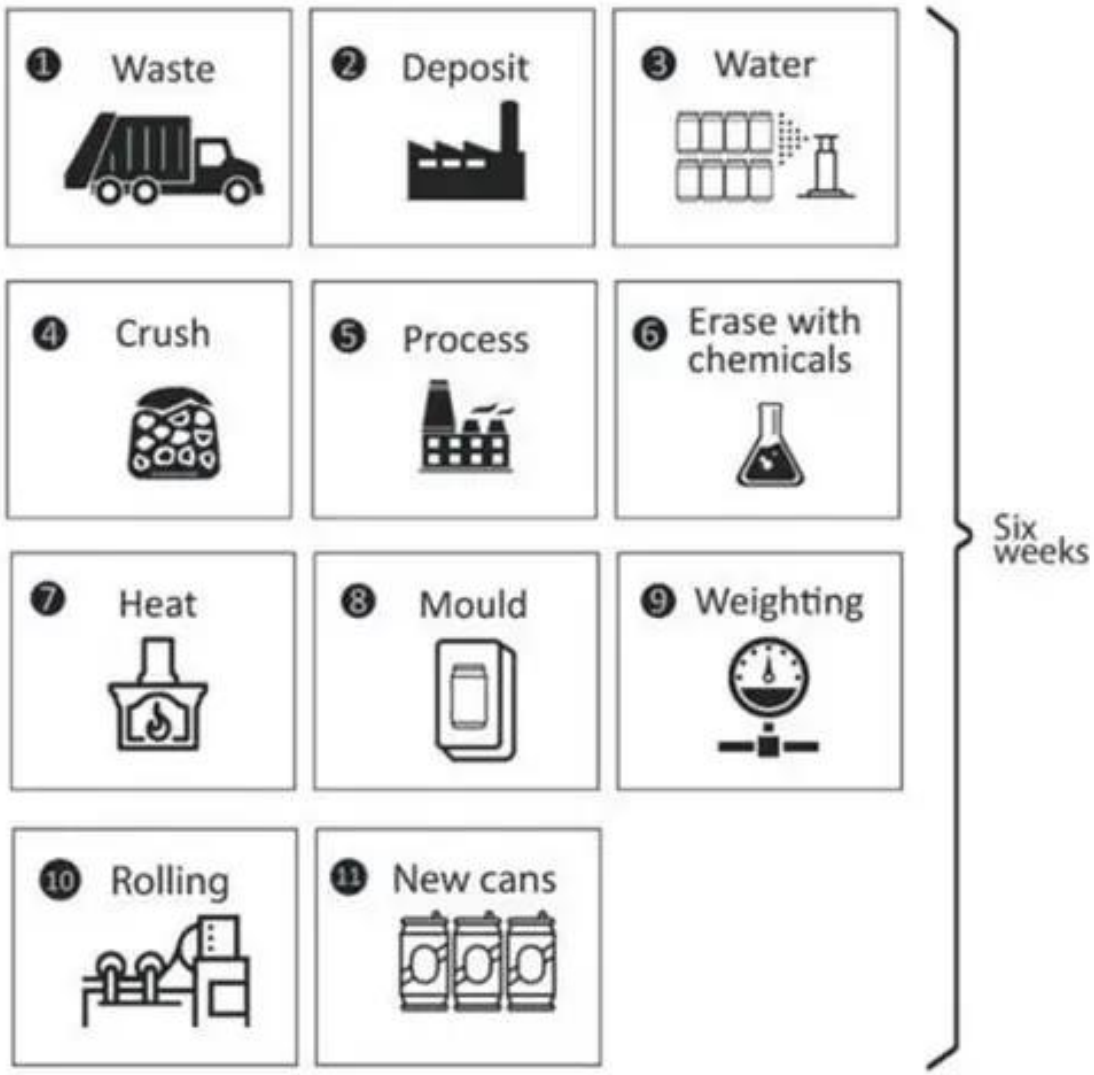
6. Now, write your own report based on the outline above.

PRACTICE 2

You should spend about 20 minutes on this task.

The diagram below shows the recycling process of aluminium cans.
Provide an overview of the information by identifying and describing the key details, and include comparisons where appropriate.

Write at least 150 words.



1. Question analysis. Answer the questions below.

a. What process does the diagram describe? _____

b. Is it natural or man-made? _____

c. Which voice should be used? active / passive

2. Complete the outline to the task 1 above, using the suggested vocabulary and / or a number.

Overview	**(cyclical / linear)** process involve ____ stage
	begin with **(waste trucks / trucks collect waste / collection of waste)**
	end with **(new cans / new cans are produced / new cans being produced)**
	take approximately ____ week to complete

When?	**What?**	**What is done?**	**Where / When / How / Why / Into what?**
When the can are **(wasted / discarded / removed),**			
During the **(assembly / treatment / dismantling)** process,			
Once the aluminum block is taken out of the **(mould / heater / roller)**			
	This completes the production **(sequence / loop)**		

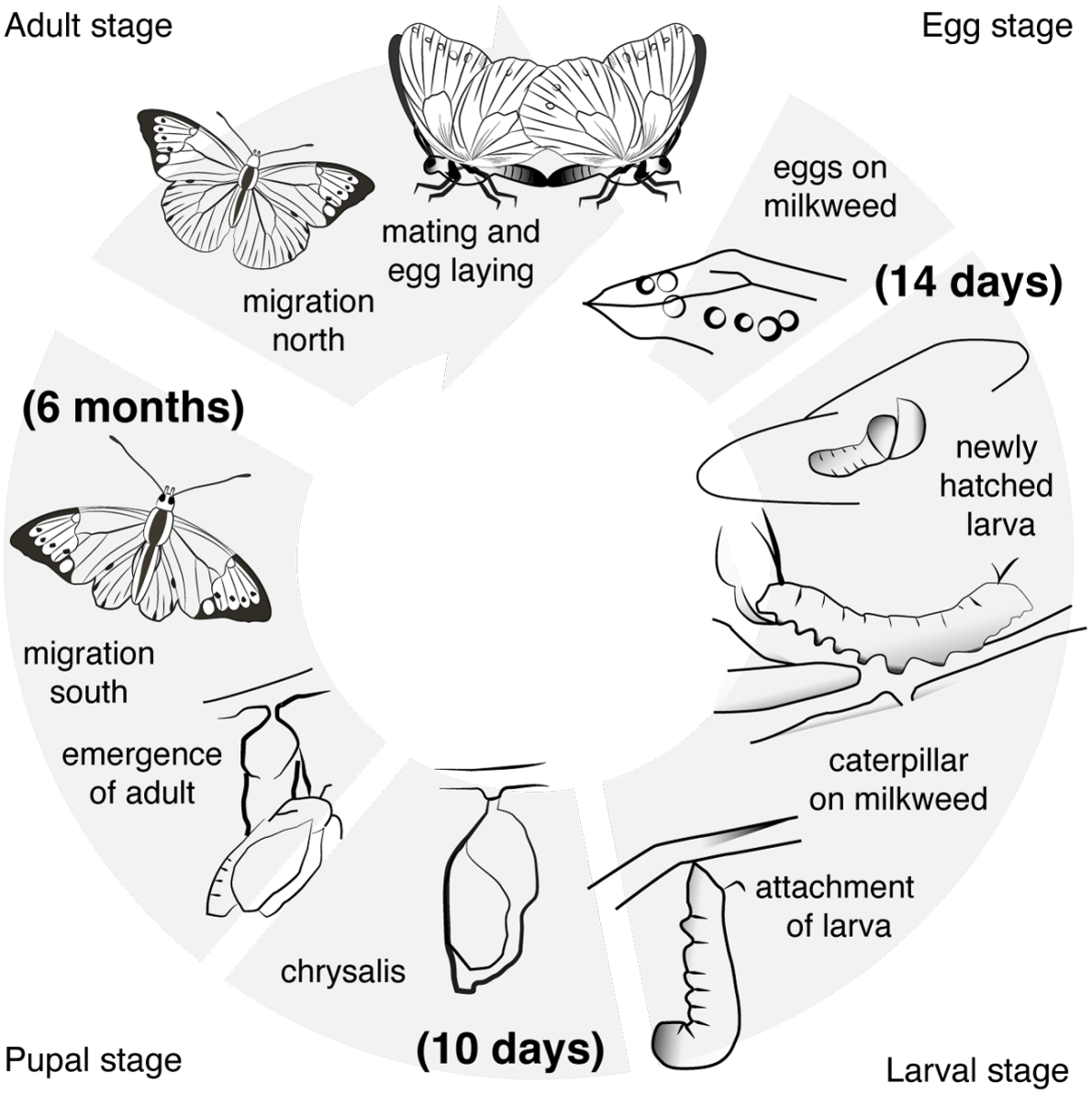
3. Now, write your own report based on the outline above.

PRACTICE 3

You should spend about 20 minutes on this task.

The diagram below shows the life cycle of the Monarch butterfly.
Summarise the information by selecting and reporting the main features, and make comparisons where relevant.

Write at least 150 words.



1. Question analysis. Answer the questions below.

d. What process does the diagram describe? _____
e. Is it natural or man-made? _____
f. Which voice should be used? active / passive

2. Complete the outline to the task 1 above, using the suggested vocabulary and / or a number.

Overview	the **(cyclical / linear)** process involves 4 main stages: _____, _____, _____, _____,
	take approximately ____ months to complete.
	combines **(physical / biological / psychological)** transformation and migration for **(seasonal / reproductive / transportation)** reasons

When?	**What?**	**What is done?**	**Where / When / How / Why / Into what?**
At the start of the life cycle,			
After ____ days of pupation period			
	This completes the natural **(sequence / cycle)**		

3. Now, write your own report based on the outline above.